

Question	Answer	Marks
3(a)(i)	(P and Q are at the) same temperature	B1
	no <u>net</u> transfer of thermal energy (between P and Q)	B1
3(a)(ii)	$Q = mc\Delta T$	C1
	$24 \times 10^3 = (0.54 \times 390 \times \Delta T) + (0.37 \times 910 \times \Delta T)$	C1
	$\Delta T = 44\text{K}$	A1
3(b)(i)	work done = $p\Delta V$	C1
	$= (1.6 \times 10^5) \times (0.18 - 0.32)$	C1
	$= -2.2 \times 10^4\text{J}$	A1
3(b)(ii)	$pV = NkT$	C1
	$N = (1.6 \times 10^5 \times 0.18) / (1.38 \times 10^{-23} \times 273)$ $= 7.6 \times 10^{24}$	A1
3(b)(iii)	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	r.m.s. speed = $\sqrt{[(3 \times 1.38 \times 10^{-23} \times (210 + 273)) / (4.7 \times 10^{-26})]}$ $= 650\text{ms}^{-1}$	A1

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